

Space Decoupled Prototype Learning for Few-Shot Attack Detection in Cyber–Physical Systems

Haili Sun , Yan Huang , *Member, IEEE*, Chunjie Zhou , Lansheng Han , Hongle Liu, Juan Chen , and Xin Li

Abstract—Due to the lack of effective attack detection measures, cyberattacks may cause strong damage to industrial cyber–physical systems (CPSs). The embedding of attack categories learned by the existing attack detection methods is highly coupled to each other with fuzzy boundaries and overlapped neighborhood, leading to weak robustness and high false positive rates. To address these issues, in this article, we propose a few-shot attack detection method based on decoupled prototype learning (DPL-FSAD), aiming to enhance the detection accuracy and generalization capabilities for malicious attacks in CPS. Specifically, we first introduce feature contrastive learning to extract differentiated features from highly similar samples, achieving compact intraclass and sparse interclass feature embedding space. To solve the problem of fuzzy boundaries of different attack categories, prototype contrastive learning is then employed to reduce the coupling degree among prototypes and enhance their discriminability. A regularization term is exploited to mitigate the overfitting problem by reducing the gap between the feature embedding and prototypes. Furthermore, an orthogonal constraint is employed to separate prototypes of different attack types, generating a decoupled prototype embedding

space. The experimental results on three public cyberattack datasets show that, compared with the suboptimal model a few-shot learning model with Siamese convolutional neural network (FSL-SCNN), the proposed DPL-FSAD can improve the precision by 5.53%, *F1*-score by 3.3%, and reduce the false positive rate by 2.37% in average, which proves that the space decoupled prototype learning is effective for improving the generalization and robustness of industrial CPS attack detection in few-shot scenario.

Index Terms—Few-shot learning (FSL), industrial cyber–physical system (CPS), network attack detection, prototype contrastive learning (PCL), space decoupling.

I. INTRODUCTION

THE rapid development of Industry 4.0 has promoted the integration of cyber network and physical systems, giving rise to a new paradigm: Cyber–physical systems (CPSs). CPSs are large, distributed, heterogeneous, and multidimensional intelligent systems, integrating communication, computation, and control to realize close combination and coordination of physical and software resources [1]. These systems provide a wide range of functionalities, including intelligent production and collaborative control, which play significant roles in advancing industrial intelligence. However, they are vulnerable to cyber–physical attacks due to the lack of effective network security defense measures [2]. Attackers may invade controllers of CPS by launching malicious code injection attacks to disrupt the normal operation of equipment and the production order of an enterprise, which may cause catastrophic consequences on the economy, environment, and life security [3], [4]. Therefore, the detection of cyberattacks against CPS is of paramount importance for timely warning, ensuring enterprise production safety, and reducing economic loss.

Attack detection aims to distinguish attack behaviors from normal system states within multivariate time series (MTS), such as network traffic, sensor signals, and system logs generated by CPS. However, due to the complexity of network attacks and the variety of network functionalities, detection technologies based on mathematical statistics, state estimation, and traditional machine learning encounter issues, such as high false alarm rate (FAR), poor adaptability, and low detection accuracy. Recently, attack detection algorithms based on deep learning have been widely explored [3], [5]. These attacks are usually sparse or even unseen in real-world scenarios, leading to insufficient labeled

Manuscript received 22 January 2024; revised 2 May 2024; accepted 18 June 2024. Date of publication 16 July 2024; date of current version 7 October 2024. The work of Lansheng Han was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3103402 and in part by the National Natural Science Foundation of China under Grant 62072200, Grant 62172176, and Grant 62127808. The work of Chunjie Zhou was supported by the National Natural Science Foundation of China under Grant 61873103, Grant 62127808, and Grant 61433006. Paper no. TII-24-0342. (Haili Sun and Yan Huang contributed equally to this work.) (Corresponding author: Lansheng Han.)

Haili Sun, Lansheng Han, Hongle Liu, Juan Chen, and Xin Li are with the Hubei Key Laboratory of Distributed System Security, Hubei Engineering Research Center on Big Data Security, School of Cyber Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: hailisun@hust.edu.cn; hanlansheng@hust.edu.cn; hongleliu@hust.edu.cn; juanchen1737@hust.edu.cn; zsgkijhy@126.com).

Yan Huang is with the National Key Laboratory of Science and Technology on Multispectral Information Processing, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: platanus@hust.edu.cn).

Chunjie Zhou is with the Key Laboratory of Image Processing and Intelligent Control, Ministry of Education, School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, China (e-mail: cjiezhou@hust.edu.cn).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TII.2024.3423327>.

Digital Object Identifier 10.1109/TII.2024.3423327

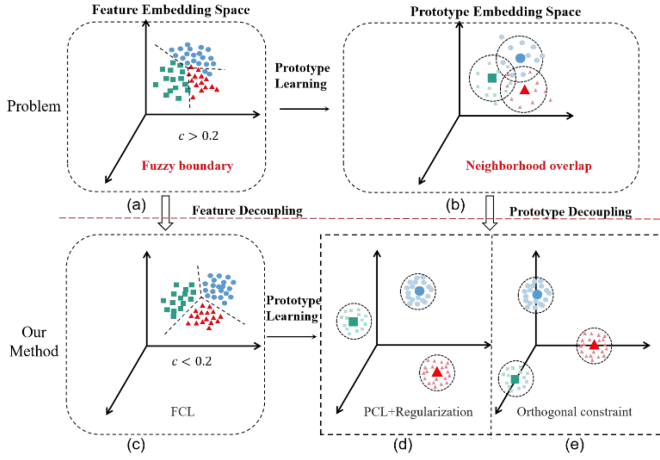


Fig. 1. Illustration of the problems and motivations. (a) Highly correlated features with large coupling degrees (e.g., $c > 0.2$) and fuzzy boundaries. (b) Overlapped prototype neighborhood. (c) Feature space decoupling based on FCL. (d) Prototype space decoupling based on PCL and regularization. (e) Decoupling via prototype orthogonal constraint.

samples for model training and problems, such as overfitting and weak generalization ability. This poses great challenges for existing attack detection methods that rely heavily on rich labeled data.

To this end, some researchers have attempted to introduce the few-shot learning (FSL) method to overcome the challenge of insufficient labeled samples. FSL enables models to effectively distinguish novel categories with few labeled samples. In recent years, several works based on FSL have been proposed for attack detection in industrial CPS. Zhou et al. [6] created a Siamese neural network for detecting industrial attacks to solve the overfitting issue. Huang et al. [7] proposed a gated network that detects different attacks by aggregating both seen and unseen types within few-shot settings.

However, due to the inherent high correlation and coupling degree of sample features between attack categories, as shown in the upper layer of Fig. 1, methods based on feature learning, such as convolutional neural networks [see Fig. 1(a)], and metalearning methods based on prototypical networks (PNs) [see Fig. 1(b)] face the problems of fuzzy boundaries and overlapped prototype neighborhood, leading to poor robustness and high FAR in the scenario of sample sparsity and few-shot cases.

Furthermore, Tian et al. [8] have demonstrated that a well-designed feature extractor is more effective than models relying solely on complex metalearning algorithm for few-shot classification tasks, as they are hard to extract more discriminative and differentiated features from industrial CPS data. To overcome the above challenges, our intuitive idea is to achieve feature space decoupling by extracting differentiated sample features and prototype space decoupling by prototypical intra-class aggregation and interclass separation so as to reduce the feature correlation and coupling degree between different attack categories, as shown in the lower layer of Fig. 1.

Therefore, this article proposes a few-shot attack detection method based on space decoupled prototype learning, called DPL-FSAD, for industrial CPS security. Specifically, to extract

distinctive features and enhance the discriminability, we first design a feature contrastive learning (FCL) [see Fig. 1(c)] module via maximizing mutual information to obtain a compact intra-class and sparse interclass feature embedding space. Second, to alleviate the problem of sample sparsity and fuzzy boundaries of different attack categories, we propose prototype contrastive learning (PCL) [see Fig. 1(d)] to reduce the coupling degree between different prototypes. In addition, the FCL and PCL may overfit in few-shot cases; thus, a robust regularization term is exploited based on the mutual information from features to prototypes. Finally, the orthogonal constraints between prototypes [see Fig. 1(e)] are introduced to make the prototype spaces of different attack categories orthogonal to each other so as to realize prototype decoupling in the latent embedding space. This not only improves the robustness of model classification but also reduces the FAR.

The main contributions of this article are as follows.

- 1) A two-stage few-shot attack detection model DPL-FSAD based on FCL and PCL is proposed to tackle the problem of high coupling degree and fuzzy boundaries between different attack types. Different attacks are spatially decoupled from the feature level and prototype level, respectively, which improves the discriminability of complex CPS attacks.
- 2) To learn more differentiated features from few labeled samples, a lightweight FCL module is designed to generate a compact intraclass and sparse interclass embedding space.
- 3) To reduce the coupling degree between different prototypes, PCL and orthogonal constraints are exploited for generating highly decoupled embedding space.
- 4) An efficient attack detection algorithm is developed based on the decoupled feature and prototype embedding space, which identifies CPS attacks with few labeled samples through the nearest neighbor in the prototype embedding space.

II. RELATED WORK

In recent years, attack detection algorithms [10], [12] have been extensively analyzed and studied for the security and reliability of industrial CPS. It is crucial to develop appropriate detection architectures to identify attacks on these systems. Several methods based on deep learning have been designed to protect CPS against cyber-physical attacks [11], such as error diagnosis [13], attack detection, and fault-tolerant control [14]. Pearce et al. [15] proposed a bidirectional runtime enforcement to mitigate the damages posed by compromised controllers in CPS. To alleviate misclassification errors, Khan et al. [27] designed a statistical feature extraction algorithm and developed an intrusion detection system based on long-short term memory network (LSTM) autoencoder to distinguish malicious actions from industrial control systems. Zhou et al. [3] proposed a variational long short-term memory model (VLSTM) for guaranteeing industry security. They designed a variational reparameterization scheme to capture the low-dimensional feature embeddings and identified anomalies by an estimation network. Wu and Guo [28] detect attacks by designing a hierarchical convolution neural

network and recurrent neural network (CNN+RNN) architecture named LuNet with a gradually increasing granularity to extract both spatial and temporal features of the data. Obviously, the aforementioned approaches show that deep-learning-based methods are promising to identify cyber-physical attacks in industrial CPS. However, as traditional supervised learning methods typically rely on rich labeled data and prior knowledge, they are difficult to detect novel attacks from few labeled samples effectively in the smart industrial environment.

Semisupervised anomaly detection: To alleviate the above problem, several researchers have developed semisupervised learning anomaly detection techniques. Huang et al. [16] developed a semisupervised learning-based framework named a semisupervised learning based active anomaly detection framework using variational auto-encoder (SLA-VAE) to detect anomalies in an industrial environment using a variational autoencoder. To detect anomaly patterns in MTS, Zheng et al. [17] integrated reconstruction model, prediction model, and auxiliary discriminator and proposed an end-to-end semisupervised anomaly detection model to make reasonable use of the limited labeled data.

Contrastive learning for attack detection: Contrast learning is widely used in computer vision [18], [19]. Due to its powerful representation learning ability, many researchers have introduced it to attack detection field [20], [21]. Kopuklu et al. [20] adopt supervised contrastive learning for detecting anomalous driving behavior. For the out-of-class detection task, Shenkar and Wolf [21] proposed to maximize the mapping relationship of mutual information between samples and their shielded part through contrastive loss. However, these methods do not incorporate the label information for temperature coefficient adjustment, which may lose some class-level features. In addition, the correlation between different categories has not been well measured.

FSL for attack detection: Few-shot learning (FSL) is a new type of transfer learning technique [22]. By reusing the transferable knowledge of existing classes, it can identify novel categories with limited supervisory information [6], [23]. Zhou et al. [6] develop an anomaly detection model named a few-shot learning model with Siamese convolutional neural network (FSL-SCNN) based on Siamese convolutional neural network to detect the anomalies in industrial CPS. Kale and Thing [24] propose an enhanced anomaly detection network based on FSL to identify anomalies using few seen class anomalies and evaluated the network on three public datasets. To optimize the resource consumption in systems, Puzanov et al. [25] propose a deep reinforcement one-shot learning framework and train a deep-Q network to obtain a policy for identifying unseen classes in testing data. To overcome the class imbalance problem, Bedi et al. [26] design a new type of intrusion system named Siam-intrusion detection systems (IDS) based on a Siamese neural network to detect attacks without using oversampling and random undersampling. However, these methods neglect the possible coupling between different classes in the learned feature space, which may lead to high false positive rates.

Different from the existing approaches, we solve the problems of fuzzy boundaries and neighborhood overlapping of different attack categories via space decoupling techniques, i.e., feature decoupling via FCL and prototype decoupling via PCL, regular-

ization, and orthogonal constraint, to achieve a high intraclass variance and low interclass coupling degree. To facilitate comparison, we also analyzed the strengths and weaknesses of the above models, as shown in Table I.

III. PROBLEM DEFINITION

In this article, the attack detection of industrial CPS is defined as an FSL task. Given a dataset D containing two general subsets D_{nor} and D_{att} indicating normal and attack samples, respectively, $D_{\text{nor}} = \{(x_{\text{nor}_i}, y_{\text{nor}_i}) | i = 1, 2, \dots, N_{\text{nor}}\}$ contains N_{nor} labeled normal samples in which x_{nor_i} is the i th normal sample and y_{nor_i} is the corresponding class label. Likewise, $D_{\text{att}} = \{(x_{\text{att}_i}, y_{\text{att}_i}) | i = 1, 2, \dots, N_{\text{att}}\}$ contains N_{att} labeled attack samples in which x_{att_i} is the attack sample and y_{att_i} is the corresponding class label. To describe the few-shot scenario, we assume that the scale of attack samples is far less than normal samples, i.e., $N_{\text{nor}} \gg N_{\text{att}}$. Thus, a set of samples from D_{att} is selected to form the support set Tr_S in each training episode, and the corresponding query set Tr_Q , which is used to indicate the unobserved samples of novel classes between different episodes, can be described as $\text{Tr}_Q = \{(x_{\text{att}_j}, y_{\text{att}_j}) | j = 1, 2, \dots, N_q\}$. Specifically, to form the N -way k -shot learning task, we randomly select N malicious attack categories from D_{att} , each category contains k samples to form the support set Tr_S and the corresponding unseen query set Tr_Q , which indicates other samples of the same categories in each training episode. The few-shot attack detection task based on prototype learning aims to learn the prototypes for each of the N attack categories using the corresponding k samples, then classify unseen samples according to the distance to the N different prototypes.

IV. METHODOLOGY

A. Framework Overview

The left part of Fig. 2 illustrates a typical architecture for attack detection in artificial intelligence (AI)-enhanced industrial CPS in which attackers may send the malicious code to compromise the CPS systems through hacking into the CANbus network or programmable logic controller (PLC) controllers. The supervisory control and data acquisition system is involved to monitor and collect sensor signals (e.g., pressure and temperature) and network flows (e.g., TX&RX packet data) generated across the cyber network, in which the AI-based attack detection module is deployed to identify malicious attacks.

Unlike the traditional classification models in industry, we formulate the detection of malicious attacks with few labeled abnormal samples in CPS as a few-shot classification task and proposed a novel and efficient attack detection architecture based on prototype learning to ensure its security and safety (upper right of Fig. 2).

Fig. 3 illustrates the architecture of the proposed DPL-FSAD, which consists of three modules, i.e., feature extractor, FCL, and decoupled prototype generation (PCL). The feature extractor is composed of two residual blocks and each block contains two convolution layers. One of the main components of the FCL module is a contrastive head with a multilayer perceptron (MLP) that maps feature into a hypersphere. In addition, the PCL

TABLE I

STRENGTHS AND WEAKNESSES OF THE METHODS (UL DENOTES UNSUPERVISED LEARNING, SSL DENOTES SEMISUPERVISED LEARNING, SL DENOTES SUPERVISED LEARNING, CL DENOTES CONTRASTIVE LEARNING, AND FSL DENOTES FEW-SHOT LEARNING)

Method	Type	Strengths and Weaknesses	Application
[11], DDLM[12], DL-AD[13]	UL	Strengths: need no labeled data, able to handle complex and high-dimensional data Weaknesses: no clear objective, sensitive to data quality	Time-series anomaly detection
SLA-VAE[16], MDAE-DT [17]	semi-SL	Strengths: handle label imbalance and label missing problems Weaknesses: need certain number of labeled data, high complexity, unable to handle inter-class coupling problem	Time-series intrusion detection
VLSTM[3],[27], LuNet[28]	SL	Strengths: accurate and reliable detection, learn temporal dependencies Weaknesses: rely heavily on labeled data, may suffer overfitting or underfitting, unable to handle interclass coupling problem	Time-series anomaly detection
[18][19][20]	Self-SL, CL	Strengths: need no labeled data, learn the intrinsic structure of data Weaknesses: unable to handle interclass coupling problem, large memory consumption, rely on data enhancement algorithm	Image anomaly detection
CLAD[21]	SL, CL	Strengths: extract distinguishable features, learn the intrinsic structure of data Weaknesses: rely heavily on labeled data, may suffer overfitting or underfitting, unable to handle inter-class coupling problem	Time-series anomaly detection
FSL-SCNN[6], HET[23], AnoS_Net[24], DeROL[25], Siam_IDS[26]	SL, FSL	Strengths: accurate and reliable detection, limited labeled samples' requirements Weaknesses: may suffer overfitting or underfitting, unable to handle inter-class coupling problem	Time-series anomaly detection
DPL-FSAD (Ours)	SL, FSL, CL	Strengths: accurate and reliable detection, limited labeled samples' requirements, learn the intrinsic structure of data, achieves high intra-class variance and low interclass coupling degree, high generalization ability, low FAR; Weaknesses: two stages training	Time-series anomaly detection

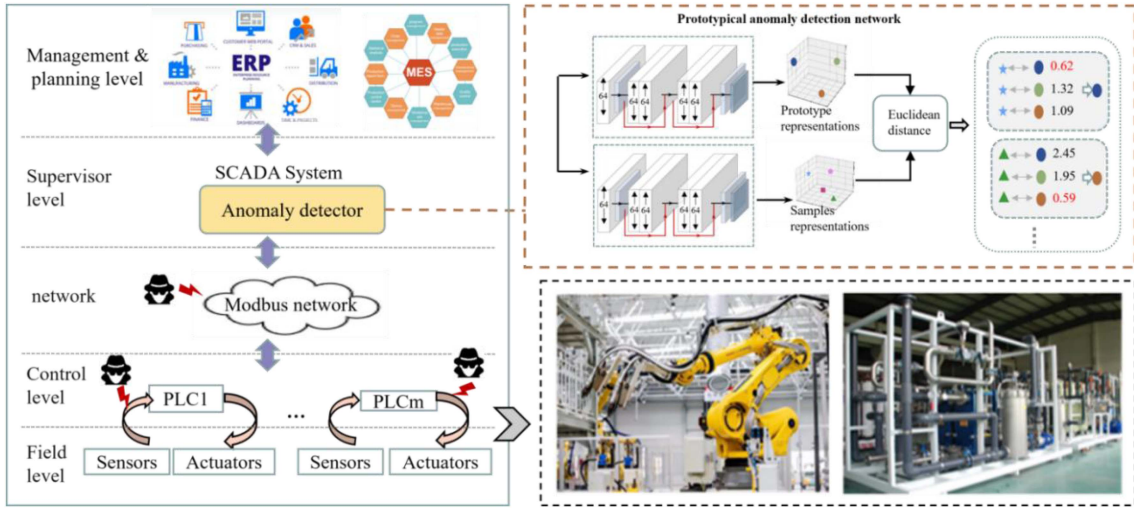


Fig. 2. Overview of a typical hierarchical layered architecture of security protection for industrial CPS.

module consists of three convolution layers with regularization constraint and orthogonal constraint to generate prototype embedding space, in which the query sample can be classified via searching for the nearest prototype.

On the one hand, the quality of features captured from input data determines the upper bound of the model. A well-designed encoder is of more significance than a complex metalearning framework in FSL [8]. Therefore, we design the specific FCL module for industrial CPS attack samples to extract differentiated high-level features (see Section IV-B for details). On the other hand, to identify unseen attacks, we formulate the attack detection process as a classification task based on class prototypes. Intuitively, the more accurate the class prototype is, the better performance the detection achieves. However, in the FSL scenario, the learned prototypes usually deviate from the true center of the class due to few labeled samples [9]. To solve

this problem, we propose decoupled PCL to make prototypes closer to the real class center and design orthogonal constraints on prototypes to separate the different prototype spaces of the attack classes, thereby reducing the coupling degree of each prototype space.

Furthermore, as industrial processes are sensitive to the false positive rate, even a minute of downtime may cause serious loss [2]. To this end, a robust regularization term is also designed based on the distance between samples and the corresponding prototypes so that similar samples will distribute closer in the feature space to generate more distinctive prototypes, which can benefit from improving the generalization ability of the model (see Section IV-C for details).

Overall, the proposed attack detection algorithm consists of two stages: the FCL stage and the PCL stage. FCL is the premise of PCL, providing it with decoupled feature representation. The

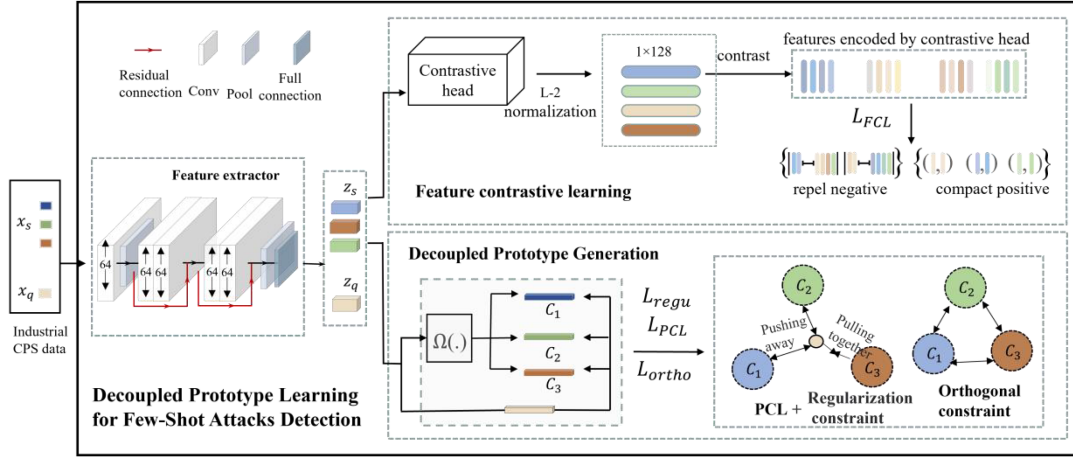


Fig. 3. Architecture of the proposed DPL-FSAD. Inputs x_s and x_q denote the samples of the support set and the query set, respectively. The input is first passed through the feature extractor to obtain the features z_s and z_q , which are then fed into the FCL and PCL modules for FCL and prototype generation tasks, respectively. In addition, in the FCL module, the contrastive head is an MLP, which encodes the features z_s and z_q into a 1×128 -dimensional vector; in the PCL module, $\Omega(\cdot)$ denotes the prototype generation function, and C_1 , C_2 , and C_3 denote the generated prototypes.

purpose of FCL is to generate a distinctive high-level feature embedding space that is compact within classes and sparse between classes. Based on the feature embedding space, the PCL aims to learn a high-quality prototype space in which prototypes are located close to the real class centers. Meanwhile, orthogonal constraints and regularization terms are designed to reduce the coupling degree of the prototype space and improve the generalization ability of the model.

B. Stage 1: FCL

Aiming at the problem that most existing attack detection methods may fail to extract the differentiated features of CPS network attacks and fuzzy boundaries, we propose the following hypothesis.

Hypothesis 1. FCL can capture common features of positive samples and differentiated features of negative samples, thus improving the significance level of key features.

According to Hypothesis 1, we first design a specific feature extractor f for CPS attacks to encode samples into an embedding space. Then, to improve the quality of the embedding space, we conduct the FCL as a supervised instance-level classification task, which is designed to recognize latent fine-grained structure in the low-dimensional feature space by separating the representations of different latent classes and aggregating those of the same latent classes simultaneously. Meanwhile, we develop a class-information injection (CII) technique for the FCL to obtain a compact intraclass and sparse interclass embedding space. The FCL process is shown in the upper right of Fig. 3.

Given input data x_i , the feature extractor f encodes it as a feature embedding r_i that can be represented as follows:

$$r_i = f(x_i, \theta) \quad (1)$$

where θ denotes the parameters of f and r_i denotes the encoded feature embedding. After that, r_i is passed to a contrastive head to conduct the following FCL process.

During the FCL process, specifically, for a minibatch of K features $\{r_i, y_i\}_{i=1}^N$, they are mapped into a hypersphere denoted as $\{z_i, y_i\}_{i=1}^N$ by the contrastive head, where $z_i = \text{MLP}(r_i) \in R^{1 \times 128}$ and $y_i \in R^{1 \times 128}$ is the label of the ground truth, MLP denotes the multilayer perceptron, then the FCL loss L_{FCL} can be defined as follows:

$$L_{\text{FCL}} = \frac{1}{N} \sum_{i=1}^N L_{z_i} \quad (2)$$

$$L_{z_i} = \frac{-1}{|N_{y_i}| - 1} \sum_{k \in N_{y_i}} \log \frac{\exp(d(z_i, z_k) / \tau)}{\sum_{q=1}^N \exp(d(z_i, z_q) / \tau)}. \quad (3)$$

Here, we denote samples with the same labels as positive samples, and samples with different labels as negative samples. N_{y_i} is the set of indices of positive samples with the same label as y_i in an episode, $|N_{y_i}|$ is its cardinality, and τ is a scalar temperature coefficient.

As in (3), $d(z_i, z_k)$ denotes the similarity between the i th and k th samples in the feature embedding space, z_k ($k \in N_{y_i}$) denotes the samples from the same cluster as z_i . The objective of L_{FCL} and L_{z_i} is to narrow the instance-level distance between samples with the same label, i.e., $d(z_i, z_k)$, and enlarge those with different labels, i.e., $d(z_i, z_q)$, $q \notin N_{y_i}$. Consequently, the samples of each category will form a compact cluster C_i with lower variance, and the boundaries between different clusters will be enlarged with a higher coupling degree. This will capture differentiated features and facilitate the following classification task to obtain more accurate class prototypes, which are evaluated in Section V-E.

Control of temperature coefficient: The temperature coefficient is used to adjust the degree of attention paid to hard samples in contrastive loss (3). The lower the temperature coefficient is, the more attention will be paid to similar samples. But for supervised contrastive learning, the category information is known. To learn the embedding space with clearer boundaries,

we design a CII technique to inject label-specific features into the FCL module so that the model will pay more attention to the category-related common features instead of the distinction between similar samples. The CII can be defined as follows:

$$\text{CII}(i, k) = \begin{cases} 1/\beta & y_i = y_k \\ 1/\tau, & y_i \neq y_k \end{cases} \quad (4)$$

where $0 < \tau < \beta$. CII assigns a lower temperature coefficient β to positive samples' pairs, resulting in the repulsion strength of the anchor toward positive samples being lower compared with negative samples. Then, the CII is multiplied by the similarity score between sample pair, so L_{z_i} in (3) can be converted to

$$L_{z_i} = \frac{-1}{|N_{y_i}| - 1} \sum_{k \in N_{y_i}} \log \frac{\exp(d(z_i, z_k)/\tau)}{\sum_{q=1}^N \exp(d(z_i, z_q) * \text{CII}(i, q))}. \quad (5)$$

By setting a small penalty rate of $1/\beta$ for the positive sample pair, the unique representations of different samples can be learned and can avoid the characteristics of positive samples from being too similar.

C. Stage 2: Decoupled Prototype Generation

Considering the prevalent issues of prototype-based FSL, in which prototypes tend to overlap and exhibit a high degree of coupling, we posit the following hypothesis.

Hypothesis 2. PCL can improve the degree of intraclass aggregation and reduce the coupling degree between prototypes.

Based on Hypothesis 2, we design a mutual information maximization (InfoMax) training objective to treat normal and attack samples as contrastive views to alleviate that problem. During the training stage, we maximize the mutual information between the same samples in query set and the corresponding prototype, which is generated based on the samples in support set, while penalizing other samples that do not belong to the prototype. In this way, samples from the query set also participate in generating prototypes. Theoretically, incorporating both the query set and the support set into the fine-tuning prototype is beneficial to obtain more accurate prototypes because the more samples involved in the prototype generation, the more accurate the prototype will be [9].

Therefore, we formalize the PCL as an InfoMax and prototype-based classification task. The proposed DPL-FSAD is forced to learn a multivariate classified F partitioning the embedding space according to the classes. For the few-shot attack detection task, introducing the multivariate classified function F cannot only alleviate the problem of inaccurate mean-based prototype computation but also simplify the whole training mechanism. That is, F can be approximated as a simple cross-entropy implementation of the InfoMax objective. Additionally, to improve the generalization ability and avoid overfitting, a regularization term and orthogonal constraints are designed to enhance the internal compactness of prototypes and reduce the coupling degree between prototypes.

Given the prototype of class i , i.e., C_i , the prototype contrastive loss can be formalized as the approximate mutual information $I(C_i, \text{Tr}_Q)$ between all the samples in query set and the

prototype vector

$$I(C_i, \text{Tr}_Q) \geq E[\log(d(C_i, x_{\text{in}}))] + E[\log(1 - d(C_i, x_{\text{not}}))] \quad (6)$$

where E denotes the expectation, and x_{in} and x_{not} denote the samples in the query set Tr_Q that in C_i or not in C_i . $d(C_i, x)$ denotes the similarity between C_i and x ; here, we choose the dot product as the metric, i.e., $d(C_i, x) = C_i \cdot x$. Then, the InfoMax loss for prototype C_i can be defined as follows:

$$L(C_i) = \frac{-1}{|Q_{\text{in}}|} \sum_{x_{\text{in}} \in Q_{\text{in}}} \log(d(C_i, x_{\text{in}})) - \frac{1}{|Q_{\text{not}}|} \sum_{x_{\text{not}} \in Q_{\text{not}}} \log(1 - d(C_i, x_{\text{not}})) \quad (7)$$

where $\{Q_{\text{in}}, Q_{\text{not}}\} \subset \text{Tr}_Q$ denotes the samples that belong to C_i and the samples that not belong to C_i , respectively. Thus, for N -way k -shot attack detection task, the final prototype contrastive loss can be defined as the sum of the InfoMax losses of the N prototypes

$$L_{\text{PCL}} = \frac{1}{N} \sum L(C_i). \quad (8)$$

Intuitively, minimizing the InfoMax loss (mutual information) is equivalent to narrowing the distance between the sample and its prototype and enlarging the distance between the sample and other prototypes, resulting in a more compact prototype embedding space with a larger margin between classes. Thus, the deviation of the prototype from the true center of the samples is reduced, refer to Section V-E for more detail.

Distance-based regularization term: Since industry (such as smart manufacturing plants) is sensitive to FAR, overfitting is usually an issue to consider in a few-shot scenario. To overcome this issue, we design a regularization term based on the distance between the sample feature embedding r and the corresponding prototype C to improve the generalization ability of the model. This distance-based regularization term is defined as follows:

$$L_{\text{regu}} = D(r, C). \quad (9)$$

L_{regu} can further improve the performance of the classifier due to the following reasons.

- 1) It enables the sample features of the same class distribute more compactly, which can implicitly enlarge the margin between different classes and, therefore, boosting the classification.
- 2) L_{PCL} emphasizes the separation of prototypes, thus by combining L_{PCL} and L_{regu} together, we can learn intraclass compact and interclass distinctive feature embeddings.

Orthogonal constraint: As shown in Fig. 1(b), the neighborhoods of prototypes learned may overlap with each other, which make it difficult to learn classification boundaries with high discriminative power. To solve this issue, we propose Hypothesis 3 as follows.

Hypothesis 3. The prototype orthogonal constraint can improve the interclass sparsity of the classifier's spatial distribution and reduce the degree of overlapping.

Algorithm 1: Training process of the proposed model DPL-FSAD

Input: A set of normal signal samples

$$D_{\text{nor}} = \{(x_{\text{nor}_i}, y_{\text{nor}_i}) | i = 1, 2, \dots, N_{\text{nor}}\}.$$

A set of attack samples

$$D_{\text{att}} = \{(x_{\text{att}_i}, y_{\text{att}_i}) | i = 1, 2, \dots, N_{\text{att}}\}$$

Output: A trained attack detection model DPL-FSAD

- 1: Initialize hyperparameter epoch E , batch size B , loss threshold ε , hyperparameter $\tau, \beta, \lambda_1, \lambda_2$
 - 2: **for** epoch = 1 to E **do**
 - 3: **for** each episode **do**
 - 4: Choose N classes with K samples from D_{nor} and D_{att} to build support set Tr_S
 - 5: Choose N classes with K samples from D_{att} to build query set Tr_Q
 - 6: **for** x_q in Tr_Q **do**
 - 7: Transform x_q into feature embedding r_q by (1)
 - 8: **for** x_s in Tr_S **do**
 - 9: Transform x_s into feature embedding r_s by (1)
 - 10: Calculate the feature contrastive loss L_{FCL} by (2) and (5)
 - 11: **for** $k=1$ to N **do**
 - 12: Calculate the prototype C_k of class k using r_s by (11)
 - 13: Calculate the prototype contrastive loss L_{PCL} based on r_q and C_k by (8)
 - 14: **for** r_j in r_q **do**
 - 15: Calculate the regularization term L_{regu} using (9)
 - 16: **for** $k=1$ to $N-1$ **do**
 - 17: Calculate the orthogonal loss L_{orth} between C_k and C_{k+1} by (10)
 - 18: Update DPL-FSAD to minimize $L_{\text{DPL-FSAD}}$ by (12)
 - 19: **if** $L_{\text{DPL-FSAD}} < \varepsilon$ **then**
 - 20: break
 - 21: **return** DPL-FSAD
-

The orthogonal constraint enforces the prototype embedding to be orthogonal to each other, that is, their inner product is approaching 0. This increases the interclass sparsity because if two prototypes are orthogonal, their projections on a dimension will not overlap. Furthermore, under orthogonal constraints, to maximize the interclass distance, the optimization algorithm will automatically adjust the prototype embedding to separate them as much as possible, thereby reducing the overlap between different categories.

According to Hypothesis 3, the correlation of different categories can be reduced by prototype space decoupling via the orthogonal constraint. To impose the orthogonal constraint, we first map the sample features to the prototype space by matrix mapping, calculate the mean of the feature maps to obtain the prototype embedding C , and then calculate the dot product between the prototypes and make it approximate to zero so that the prototype spaces are orthogonal to each other. The prototype orthogonal loss function is defined as follows:

$$L_{\text{orth}}(M) = \sum_{i,j \in [1,N], i \neq j} C_i \cdot C_j \quad (10)$$

where M is a matrix to map each sample from feature embedding space to prototype embedding space and trained according to the prototype orthogonal constraint.

Prototype generation: For the proposed model, we generate prototypes based on feature mapping and average operations. Given a minibatch of k features $\{r_i, i \in [1, k]\}$, the support set Tr_S , the feature extractor first embeds all samples into the latent feature space. Then, feature embedding of the same class is mapped by a matrix M and aggregated into a prototype C in the prototype embedding space. Typically, it is calculated as an average of those embedding

$$C = \Omega(\cdot) = \frac{1}{k} \sum_{i=1}^k M * r_i \quad (11)$$

where k denotes the number of samples per class in Tr_S and r_i is the feature embedding of the i th sample.

D. Learning Strategy

To achieve efficient training performance, the overall loss of the proposed DPL-FSAD model is a combination of the four losses introduced above, which can be formulated as follows:

$$L_{\text{DPL-FSAD}} = L_{\text{FCL}} + L_{\text{PCL}} + \lambda_1 L_{\text{regu}} + \lambda_2 L_{\text{orth}} \quad (12)$$

where λ_1 and λ_2 are the coefficients to balance the loss of regularization and orthogonal terms, respectively. The training process of the proposed model DPL-FSAD is shown in Algorithm 1, and the parameters are learned via minimizing $L_{\text{DPL-FSAD}}$.

E. Attack Detection With DPL-FSAD

During the test stage, given the learned prototype embedding, the model first extracts the feature of the test samples and then predicts the category of the sample based on the similarity between the feature embedding and each prototype.

In detail, we measure the similarity of samples and prototypes by the distance between them. Thus, the probability $p(x \in y_i | x)$ that x belongs to the prototype C_i should be proportional to the negative of the distance between them. To satisfy the property that probability must be normalized and nonnegative, we define $p(x \in y_i | x)$ as follows:

$$p(x \in y_i | x) = \frac{\exp(-D(f(x), C_i))}{\sum_j \exp(-D(f(x), C_j))} \quad (13)$$

where $D(f(x), C_i) = \|f(x) - C_i\|_2^2$ represents the Euclidean distance between sample x and class y_i . Based on this probability, the overall prediction error can be measured via the negative log probability

$$L_{\text{pred}} = -\log p(x \in y_i | x). \quad (14)$$

The attack type of the input sample x can be identified via the closest prototype with the highest probability $p(x \in y_i | x)$.

V. EXPERIMENTS AND ANALYSIS

A. Experimental Settings

Datasets: To verify the effectiveness of our proposed model on attack detection, we conducted a plenty of experiments on

TABLE II
STATISTICS OF THE THREE PUBLIC DATASETS

Dataset	Train	Test	Feature	Attack types
TON_IoT	421 043		43	Scanning, Dos, DDos, Ransomware, Backdoor, Injection, Cross-site Scripting (XSS), Password, Man-in-the-middle
UNSW-NB15	175 341	82 332	49	Backdoors, Generic, Analysis, Shellcode, Dos, Fuzzers, Worms, Reconnaissance, Exploits
NSL-KDD	125 973	22 544	41	Dos Attack, Probe (Probing Attack), R2L Attack, U2R Attack
ERENO IEC-61850	2 955 737	2 955 647	69	Random Replay, Inverse Replay, Masquerade Attacks Outage, Masquerade Attacks Equipment Damage, Random Message Injection, High-Status Number, Poisoned High-Rate

four public datasets, i.e., UNSW-NB15 [29], a network security detection dataset upgraded from the KDD CUP99 dataset (NSL-KDD) [30], TON_IoT [31], and ERENO IEC-61850 [32]. The details of the four datasets are shown in Table II, which were preprocessed accordingly for few-shot settings.

UNSW-NB15 was created by the Intelligent Security Group of Australia and widely used to develop approaches in different systems, such as the Internet of Things (IoT), SCADA, and Industrial IoT. The IXIA PerfectStorm tool is utilized as an attack traffic generator along with normal traffic, and the attack behavior is nourished from the CVE site for the purpose of a real representation of a modern threat environment. These attack behaviors contain nine types of network attacks that may result in data leakage, tampering, illegal control, or service interruption of CPS systems.

NSL-KDD is an upgraded version of the KDD CUP99 [30] dataset, addressing the inherent issues by incorporating data from the DARPA '98 IDS Evaluation Program. It utilizes the same environment (U.S. Air Force LAN), and the simulation ended with 41 features for each connection along with the class label using the Bro-IDS tool. NSL-KDD offers a comprehensive benchmark for intrusion detection research. Notably, both training and testing sets contain four simulated attacks, including denial of service (Dos), probe, remote-to-local (R2L), and user-to-root (U2R).

TON_IoT is a new generation Industry 4.0 and Industrial IoT dataset collected from a realistic and large-scale network designed at the Cyber Range and IoT Labs, UNSW Canberra @ the Australian Defence Force Academy. The whole dataset covers nine attack families, including heterogeneous data sources collected from telemetry datasets of IoT and IIoT sensors, operating systems datasets of Windows 7 and 10, as well as Ubuntu 14 and 18 TLS and network traffic datasets. The datasets were gathered in parallel processing to collect several normal and cyberattack events from network traffic, Windows audit traces, Linux audit traces, and telemetry data of IoT services.

ERENO IEC-61850 is a simulation of an existing Brazilian transmission line and generated more than 4.4 GB of data containing seven attack types that target the electric grid substations networks based on the IEC-61850 standards. These attacks may

compromise the IEC-61850 communication protocols and cause improper functioning of the power system. It was collected by Quincozes et al. [32] via simulating an electric grid substation network traffic to produce representative features using an extensible tool ERENO.

Comparative models: To validate the advancement of DPL-FSAD, four FSL methods (i.e., FSL-SCNN [6], AnoS_Net [24], DeROL[25], and Siam_IDS [26]), three deep-learning-based methods (i.e., VLSTM [3], AE_LSTM [27], and LuNet [32]), and two semisupervised learning methods (i.e., SLA-VAE [16] and MDAE-DT [17]) were selected for comparison.

Implementation details: The classifier of the model is implemented based on the PN. For the N -way k -shot attack detection task, we set $k = \{1, 5, 10\}$ in all experiments and select the best results. The ratio of samples in support set Tr_S and query set Tr_Q is set to 1:1. We set the initial learning rate to 0.001, choose stochastic gradient descent as the optimizer, and iterate for 1000 epochs to train the model. Empirically, we choose λ_1 and λ_2 from $\{0.1, 0.05, 0.01, 0.005, 0.001\}$ and set $\lambda_1 = 0.001$ and $\lambda_2 = 0.005$ for the best performance.

Evaluation metrics: We apply and calculate four widely used metrics, precision, recall, $F1$, and FAR depending on whether normal and abnormal samples have been recognized correctly or not, to prove the detection performances of the above methods. Besides, we introduce the following two metrics, i.e., variance σ and coupling degree c to measure the degree of intraclass aggregation and coupling degree between different categories, respectively, in the embedding space

$$\sigma_i = \text{var}(C_i) \quad (15)$$

$$c = \frac{|\text{cov}(C_i, C_j)|}{\sqrt{\sigma_i} \times \sqrt{\sigma_j}} \quad (16)$$

where $\text{var}(C_i)$ denotes the variance of the class i , and $\text{cov}(C_i, C_j)$ denotes the covariance of the classes i and j . The range of c is $[0, 1]$, namely, the larger the value of c , the higher the coupling degree between two classes.

Feature selection: To avoid overfitting and curse of dimensionality, we performed feature selection before training by using the SelectFromModel interface of sklearn, which selects features by assessing their importance to the model performance. Since this article focuses on the classification task, we used SVM.LinearSVC as the sparse estimator and $L1$ norm as the penalty term for feature selection. With this setting, the results after feature selection for these four datasets were UNSW-NB15 with 31 features, NSL-KDD with 26 features, TON_IoT with 25 features, and ERENO IEC-61850 with 32 features.

B. Attack Detection Evaluation

Robustness evaluation: To evaluate the robustness of our proposed DPL-FSAD method on detecting cyberattacks, we conduct comparative experiments with nine models on UNSW-NB15, NSL-KDD, TON_IoT, and ERENO IEC-61850 datasets. Table III illustrates the remarkable detection capabilities of DPL-FSAD on the four datasets, surpassing previously selected benchmark models. Impressively, DPL-FSAD achieves the highest precision, recall, $F1$ -score, and FAR on all four

TABLE III
STATICS OF THE FOUR PUBLIC DATASETS

Datasets Metrics		UNSW-NB15					NSL-KDD				TON IoT				ERENO IEC-61850			
		pre	rec	F1	FAR	pre	rec	F1	FAR	pre	rec	F1	FAR	pre	rec	F1	FAR	
Supervised Methods	LuNet	81.2	97.3	88.5	24.4	89.2	90.8	90.1	22.0	80.3	94.6	87.4	19.5	78.5	90.5	84.1	24.9	
	AE-LSTM	73.3	97.4	83.7	28.1	80.4	86.2	83.2	35.1	82.5	83.7	83.1	35.3	79.6	80.2	79.9	40.5	
	VLSTM	86.0	97.8	90.7	11.7	85.5	96.3	91.2	12.3	91.0	97.1	94.3	13.2	91.0	92.9	91.9	13.2	
Semisupervised Methods	SLA-VAE	92.2	96.8	94.4	5.5	92.8	96.5	94.6	4.9	94.2	97.1	95.6	7.2	90.5	92.1	91.3	13.9	
	MDAE-DT	85.8	96.5	90.8	12.5	83.6	92.4	87.8	13.4	84.3	94.6	89.2	15.6	85.3	92.0	88.5	20.7	
FSL methods	DeRoL	87.1	87.0	87.0	25.7	85.7	84.6	85.1	24.3	92.2	85.0	88.5	26.8	89.6	83.2	86.3	31.2	
	AnoS_Net	73.3	96.3	83.2	33.0	83.1	86.8	84.9	28.6	78.4	92.2	85.2	34.2	79.8	90.4	84.8	39.5	
	Siam_IDS	80.7	95.6	87.2	26.6	86.1	90.5	88.3	23.2	84.3	96.8	90.5	18.5	83.1	92.7	87.6	23.2	
	FSL-SCNN	90.6	96.8	93.6	5.7	90.4	96.4	93.3	5.4	93.4	96.9	95.1	7.5	91.4	92.8	92.1	12.2	
DPL-FSAD (Our Method)	$\lambda_1=0.1$	97.3	96.0	95.8	4.5	96.9	95.8	96.3	5.0	97.0	97.4	97.2	5.2	96.1	95.5	95.8	6.3	
	$\lambda_1=0.01$	96.8	95.0	95.9	4.0	97.0	95.1	96.0	4.4	96.8	97.7	97.2	4.9	94.7	95.6	95.1	7.2	
	$\lambda_1=0.001$	96.5	97.4	96.7	3.5	97.2	97.7	97.5	3.7	97.3	98.4	97.8	4.2	95.8	96.9	96.3	5.4	
	$\lambda_1=0.0001$	95.2	96.5	94.7	8.0	95.8	96.5	96.1	8.8	96.2	96.3	96.2	7.9	93.5	94.2	93.8	9.8	
	Improvement	14.3	10.4	12.3	12.0	14.4	11.2	12.9	11.2	13.1	11.3	12.2	13.0	14.4	14.1	114.2	16.8	

Bold indicates the optimal result and underline indicates the suboptimal result. Improvement is obtained by comparing the result of DPL-FSAD with that of the suboptimal model when $\lambda_1 = 0.001$.

datasets. On UNSW-NB15 dataset, although achieving comparable performance in the recall metric with VLSTM, DPL-FSAD outperforms the suboptimal model SLA-VAE by a significant margin, marking an absolute improvement of 4.3% in precision, 2.3% in $F1$ -score, and a reduction of 2.0% in FAR. On *NSL-KDD* dataset, DPL-FSAD is significantly better than other comparison models. Compared with the suboptimal model SLA-VAE, DPL-FSAD showcases an absolute improvement of 4.4% in precision and 2.9% in the $F1$ -score. Moreover, our model exhibits a notable reduction of 1.2% in FAR. On *TON_IoT* and *ERENO IEC-61850* datasets, compared with the suboptimal model, our DPL-FSAD achieves an average improvement of 3.75% in precision, 2.7% in recall, 3.2% in $F1$ -score, and a reduction of 4.9% in FAR, implying the effectiveness of the DPL-FSAD in industrial CPS traffic attack detection. Overall, DPL-FSAD outperforms the competing methods across all evaluated metrics, demonstrating its superior robustness in detecting seen attacks.

Generalization evaluation: To verify the generalization ability of DPL-FSAD for detecting unseen attacks, taking the UNSW-NB15 dataset as an example, we randomly excluded two attacks (i.e., Analysis and Worms) from the training set and retained both attacks in the testing set. Then, the two sets were used to train and test the model, respectively. As a result, an $F1$ -score of 94.5% was obtained on detecting these two types of attacks, which demonstrates the generalization ability of DPL-FSAD.

C. Impact of Regularization Factor

As the proposed regularization term enhances the cohesion of the prototypes, a higher level of cohesion leads to improved model performance. Hence, we conducted additional experiments to examine the impact of the regularization factor on model performance, as presented in Table III. The results demonstrate that the model's performance consistently improves with decreasing values of the regularization factor. It reaches its optimal point at 0.001 and, subsequently, declines as the regularization factor decreases further. This observation suggests that there exists a nonlinear relationship between prototype cohesion and the regularization term, which warrants future investigation.

D. Ablation Study

Ablation experiments are carried out on UNSW-NB15, NSL-KDD, TON_IoT, and ERENO IEC-61850 datasets to verify the

necessity and effectiveness of each module of the proposed model in identifying attacks. We compare five variants of the proposed model:

- 1) DPL-FSAD (ours);
- 2) w/o FCL: without feature contrastive learning;
- 3) w/o PCL: without prototype contrastive learning;
- 4) w/o Regu: without regularization constraint;
- 5) w/o Ortho: without orthogonal constraint.

Table IV presents the results of ablation experiments. As can be seen from the table, each of our variants can effectively help the model to obtain better performance. Compared with the FSL-SCNN model from Table III, we see that all variants of DPL-FSAD can outperform FSL-SCNN except for w/o Regu, which indicate that the regularization term is essential for high performance. From the metrics of variance σ and coupling degree c , we can find that the absence of the regularization term leads to an increase in the variance and coupling degree; this exactly explains the decrease in attack detection performance.

In addition, all these variants without FCL, PCL, and orthogonal constraint show a decrease in $F1$ and an increase in FAR, as well as an increase in σ and c . It proves that these modules are necessary and can promote the performance of the model. In these three variants, the absence of PCL module poses the biggest impact on model performance, and the increase of σ and c demonstrates that the reason for the performance degradation may be that the intraclass dispersion and interclass coupling become stronger.

In summary, the ablation results confirm the remarkable capabilities of our proposed DPL-FSAD model. It excels in accurately learning distinctive features, reducing the coupling degree of prototypes, and delivers impressive performance enhancements, particularly in scenarios with few labeled samples. These findings underscore the effectiveness and potential of our model in the field of attack detection in industrial CPS.

E. Hypothesis Evaluation

To demonstrate the validity of the three hypotheses, as proposed in Section IV, we measured the degree of intraclass aggregation and the degree of interclass dispersion by the variance metric σ and the coupling degree c in Table III.

As shown, without the FCL module, the variance σ and the coupling degree c increased by 15.5% and 100.5% on average,

TABLE IV
RESULTS OF ABLATION STUDY ($F1$ AND FAR ARE MEASURED IN PERCENTAGES)

	UNSW-NB15				NSL-KDD				TON_IoT				ERENO IEC-61850			
	$F1$	FAR	σ	c	$F1$	FAR	σ	c	$F1$	FAR	σ	c	$F1$	FAR	σ	c
DPL-FSAD	96.74	3.5	0.36	0.134	97.54	3.7	0.34	0.127	97.80	4.2	0.33	0.125	96.34	5.4	0.41	0.156
w./o FCL	94.37	3.9	0.44	0.256	93.52	4.0	0.37	0.273	96.83	4.5	0.38	0.245	94.62	5.8	0.55	0.264
w./o PCL	93.25	4.2	0.47	0.281	93.37	4.3	0.44	0.265	96.35	4.6	0.43	0.268	93.19	6.1	0.59	0.283
w./o Regu	92.41	3.8	0.58	0.365	92.91	3.9	0.55	0.358	93.78	4.4	0.54	0.362	92.58	6.2	0.62	0.375
w./o Ortho	96.25	4.0	0.46	0.165	96.04	4.1	0.41	0.159	96.64	4.3	0.38	0.154	95.69	6.0	0.60	0.188

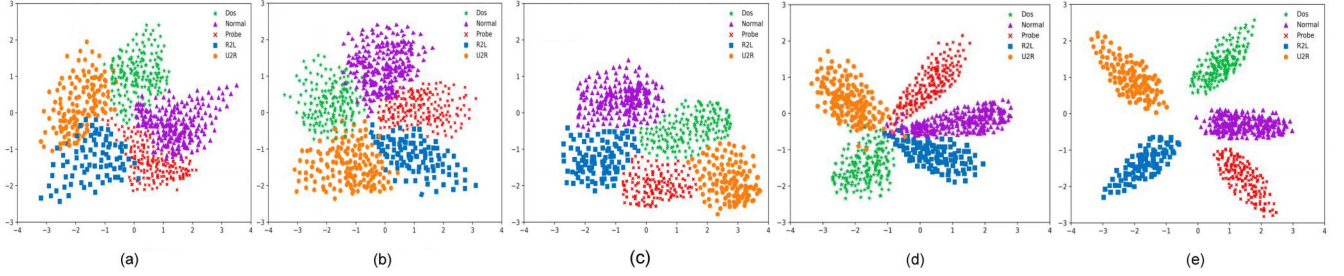


Fig. 4. Feature embedding visualization based on PCA. (a) AnoS_Net. (b) DeROL. (c) Siam_IDS. (d) FSL-SCNN. (e) DPL-FSAD.

which proves that FCL is helpful to reduce the coupling degree and variance. The lower variance indicates that the extracted features are more significant, and the lower coupling degree indicates that the features are more distinctive, these confirm the feasibility of Hypothesis 1.

Second, when without the PCL module, the variance σ and coupling degree c increased by 30.1% and 110.9% on average, which proves that PCL can significantly reduce the degree of intraclass aggregation and the interclass coupling degree (Hypothesis 2), and the effect is more significant than that of FCL module.

Finally, compared with the variant without Ortho, the variance σ and coupling degree c of the proposed model reduced by 21.4% and 23.8% on average, which indicates that the orthogonal constraint on the prototypes can improve the sparsity of the spatial distribution; it can also promote the degree of intraclass aggregation; the lower variance and coupling degree are helpful to mitigate the overlapping of neighborhoods (Hypothesis 3).

F. Feature Visualization

To further verify the decoupling effect of the proposed model, the feature representations of the compared FSL methods are visualized by performing dimensionality reduction via PCA on the NSL-KDD dataset, as shown in Fig. 4. The average intraclass variance and interclass coupling of the four models are as follows: AnoS-Net ($\sigma = 0.45$ and $c = 0.332$), DeROL ($\sigma = 0.44$ and $c = 0.325$), Siam_IDS ($\sigma = 0.43$ and $c = 0.319$), FSL-SCNN ($\sigma = 0.38$ and $c = 0.207$), and DPL-PSAD ($\sigma = 0.34$ and $c = 0.127$). It demonstrates that the proposed model DPL-FSAD can effectively reduce the variance of the intraclass distribution (by 10.5% compared with FSL-SCNN). In addition, it reduces the coupling degree between prototypes (38.6% lower than FSL-SCNN), thereby making different categories sparsely

TABLE V
INFERENCE TIME (S) OF THE FOUR FSL ATTACK DETECTION METHODS ON UNSW-NB15 DATASET

Method	UNSW-NB15	NSL-KDD
DeROL	0.3289	0.2724
AnoS_Net	0.3158	0.2512
Siam_IDS	0.3297	0.2657
FSL-SCNN	0.3165	0.2453
DPL-FSAD	0.2935	0.2349

distributed, this is beneficial to distinguish between different attack types.

G. Complexity Analysis

As most terminal devices in CPS are resource constrained, high performance and low overhead are usually priorities; computational complexity is an important indicator to measure whether a detection model is practical for integration into real-time IDS systems. To evaluate the computational complexity of the model, we calculated the inference time (taking the average of 100 experiments) of the proposed model and four few-shot detection models on the UNSW-NB15 and NSL-KDD datasets, as shown in Table V. It can be seen that the proposed model can quickly complete the detection task, and the inference time on both datasets is better than the other four methods. This observation indicates that our method is more suitable for deployment in resource-constrained CPS devices.

VI. CONCLUSION

To ensure the network security of industrial CPSs, this article proposed a new model DPL-FSAD to address the problems of existing attack detection methods with high coupling degrees and fuzzy boundaries between attack categories. Comparative experiments as well as ablation studies were conducted with

existing advanced attack detection algorithms and several traditional machine learning methods on TON_IoT, UNSW-NB15, and NSL-KDD in few-shot settings. The experimental results showed that our proposed method outperforms the existing models in terms of precision, recall, $F1$ -score, and FAR metrics. It validated the effectiveness of the three hypotheses and demonstrated the ability of DPL-FSAD for extracting distinctive attack features, space decoupling of different prototypes, and mitigating the fuzzy boundaries, which contribute to the robust detection of attack signals in industrial CPS environment in few-shot cases.

APPENDIX

Proof of Hypothesis 1.

- 1) Since $\log$ is an increasing function, to minimize L_{z_i} in formula (3), we need to maximize $\log \frac{\exp(D(z_i, z_k)/\tau)}{\sum_{q=1}^N \exp(D(z_i, z_q)/\tau)}$, i.e., maximize the numerator term $\exp(D(z_i, z_k)/\tau)$ and minimize the denominator term $\sum_{q=1}^N \exp(D(z_i, z_q)/\tau)$

$$\begin{aligned} \max \{ \exp(D(z_i, z_k)/\tau) \} &\Rightarrow \max \{ D(z_i, z_k) \} \\ &= \max \{ z_i \cdot z_k \} = \max \{ |z_i| |z_k| \cos \langle z_i, z_k \rangle \} \\ \min \left\{ \sum_{q=1}^N \exp(D(z_i, z_q)/\tau) \right\} &\Rightarrow \min \{ D(z_i, z_q) \} \\ &= \min \{ z_i \cdot z_q \} = \min \{ |z_i| |z_q| \cos \langle z_i, z_q \rangle \}. \end{aligned}$$

Under normalization condition, $|z_i| = |z_k| = |z_q| = 1$, $\max \{ |z_i| |z_k| \cos \langle z_i, z_k \rangle \} = \max \{ \cos \langle z_i, z_k \rangle \}$, i.e., to maximize the similarity between the features of in-class samples and capture the common features; $\min \{ |z_i| |z_q| \cos \langle z_i, z_q \rangle \} = \min \{ \cos \langle z_i, z_q \rangle \} = \max \{ 1 - \cos \langle z_i, z_q \rangle \}$, i.e., to minimize the similarity and maximize the distance between the features of different classes of samples. Therefore, for negative samples, the model is more inclined to capture differentiated features that are distinguishable from positive samples.

- 2) The significance level of a feature can be represented by the eigenvalue λ corresponding to the feature vector.

As $z_i \cdot z_k \leq \frac{z_i \cdot z_i^T + z_k \cdot z_k^T}{2}$, $\max \{ z_i \cdot z_k \} \Rightarrow \max \{ \sup(z_i \cdot z_i^T + z_k \cdot z_k^T) \} \Rightarrow \max \{ z_i \cdot z_i^T \}$ and $\max \{ z_k \cdot z_k^T \}$, z_i and z_k belong to the same category.

As $z_i \cdot z_i^T$ is a square matrix, $\max \{ z_i \cdot z_i^T \} \Rightarrow \max \{ \det(z_i \cdot z_i^T) \} \Rightarrow \max \{ \prod_{j=1}^n \lambda_{i_j} \}$.

Similarly, $\max \{ z_k \cdot z_k^T \} \Rightarrow \max \{ \prod_{j=1}^n \lambda_{k_j} \}$, i.e., the positive sample eigenvalues can be maximized via orthogonal constraints.

In addition, as $z_i \cdot z_q \geq -\frac{z_i \cdot z_i^T + z_q \cdot z_q^T}{2}$, z_i and z_q belong to different categories

$$\begin{aligned} \min \{ z_i \cdot z_q \} &\Rightarrow \max \{ \inf(z_i \cdot z_i^T + z_q \cdot z_q^T) \} \\ &\Rightarrow \max \{ z_i \cdot z_i^T \} \text{ and } \max \{ z_q \cdot z_q^T \}. \end{aligned}$$

As $z_q \cdot z_q^T$ is a square matrix, $\max \{ z_q \cdot z_q^T \} \Rightarrow \max \{ \det(z_q \cdot z_q^T) \} \Rightarrow \max \{ \prod_{j=1}^n \lambda_{q_j} \}$, i.e., to maximize the eigenvalues of negative samples feature vectors.

Therefore, through FCL, the significance level of the key features can be improved by maximizing the eigenvalues of positive samples and negative samples.

Proof of Hypothesis 2.

- 1) Intra-class aggregation can be measured by the average expectation of the similarity between intra-class samples and different class prototypes. Based on the prototype contrastive loss defined in formula (7), since both the left and right terms of the loss are positive, to minimize $L(C_i)$, the left and right terms need to be minimized separately

$$\begin{aligned} &\min \left\{ \frac{-1}{|Q_{in}|} \sum_{x_{in} \in Q_{in}} \log(d(C_i, x_{in})) \right\} \\ &\Rightarrow \max \left\{ \sum_{x_{in} \in Q_{in}} \log(d(C_i, x_{in})) \right\} \\ &\Rightarrow \max \left\{ \sum_{x_{in} \in Q_{in}} d(C_i, x_{in}) \right\} \\ &\Rightarrow \max \left\{ \sum_{x_{in} \in Q_{in}} C_i \cdot x_{in} \right\} \\ &\Rightarrow \max \left\{ C_i \cdot \sum_{x_{in} \in Q_{in}} x_{in} \right\} \Rightarrow \max \{ C_i \cdot E(x_{in}) \}. \end{aligned}$$

Therefore, for the cluster of prototype C_i , we only need to maximize the similarity between the sample expectation $E(x_{in})$ and the prototype C_i , i.e., to maximize the degree of intra-class aggregation.

- 2) Prototype coupling degree can be measured by formula (16), where C_i and C_j denote the prototypes of different categories, respectively, c denotes the coupling degree of prototypes, $c \in [0, 1]$, $c = 1$ exist only if $i = j$.

If $i \neq j$, to minimize the right term of formula (7)

$$\begin{aligned} \min L(C_i) &\Rightarrow \min \left\{ -\frac{1}{|Q_{not}|} \sum_{x_{not} \in Q_{not}} \log(1 - d(C_i, x_{not})) \right\} \\ &\Rightarrow \max \left\{ \frac{1}{|Q_{not}|} \sum_{x_{not} \in Q_{not}} \log(1 - d(C_i, x_{not})) \right\} \\ &\Rightarrow \max \left\{ \frac{1}{|Q_{not}|} \sum_{x_{not} \in Q_{not}} (1 - d(C_i, x_{not})) \right\} \\ &\Rightarrow \min \left\{ \frac{1}{|Q_{not}|} \sum_{x_{not} \in Q_{not}} d(C_i, x_{not}) \right\} \\ &\Rightarrow \min \left\{ \frac{1}{|Q_{not}|} \sum_{x_{not} \in Q_{not}} C_i \cdot x_{not} \right\}. \end{aligned}$$

For any $C_j = \frac{C_i}{|Q_j|} \sum_{x_j \in Q_j} x_j \in Q_{\text{not}}$, where $i \neq j$, $\min L(C_i) \Rightarrow \min\{C_i \cdot C_j\} \Rightarrow \min\{c\}$, i.e., to reduce the coupling degree between prototypes.

Proof of Hypothesis 3.

1) To minimize $L_{\text{orth}}(M)$ in formula (10), we have

$$\begin{aligned} \min\{L_{\text{orth}}(M)\} &\Rightarrow \min \left\{ \sum_{i,j \in [1,N], i \neq j} C_i \cdot C_j \right\} \\ &\Rightarrow \max \left\{ \sum_{i,j \in [1,N], i \neq j} (1 - C_i \cdot C_j) \right\} \end{aligned}$$

i.e., the similarity of different types of prototypes is minimized.

As the intraclass samples are aggregated near the prototypes, i.e., $d(C_i, x_{\text{in}}) \rightarrow 0$; therefore, the sparsity between different classes increases.

2) Interclass overlapping can be measured by the distance between samples of different classes

$$\begin{aligned} \min\{L_{\text{orth}}(M)\} &\Rightarrow \min \left\{ \sum_{i,j \in [1,N], i \neq j} C_i \cdot C_j \right\} \\ &\Rightarrow \min \left\{ \left(\frac{1}{k_i} \sum_{i=1}^{k_i} M * r_i \right) \cdot \left(\frac{1}{k_j} \sum_{j=1}^{k_j} M * r_j \right) \right\} \\ &\Rightarrow \min \left\{ \left(\sum_{i=1}^{k_i} r_i \right) \cdot \left(\sum_{j=1}^{k_j} r_j \right) \right\} \\ &\Rightarrow \min \left\{ \left(\sum_{i=1}^{k_i} \sum_{j=1}^{k_j} r_i \cdot r_j \right) \right\} \\ &\Rightarrow \max \left\{ \sum_{i=1}^{k_i} \sum_{j=1}^{k_j} (1 - r_i \cdot r_j) \right\}. \end{aligned}$$

The similarity between different types of samples is minimized, i.e., the distance between different types of samples is maximized; thus, the degree of overlapping between different classes is reduced.

REFERENCES

- [1] L. Hou, Y. Li, W. Luo, and H. Sun, "Adaptive tracking control of switched cyber-physical systems with cyberattacks," *Appl. Math. Comput.*, vol. 415, 2022, Art. no. 126721.
- [2] A. Humayed, J. Lin, F. Li, and B. Luo, "Cyber-physical systems security—A survey," *IEEE Internet Things J.*, vol. 4, no. 6, pp. 1802–1831, Dec. 2017.
- [3] X. Zhou, Y. Hu, W. Liang, J. Ma, and Q. Jin, "Variational LSTM enhanced anomaly detection for industrial big data," *IEEE Trans. Ind. Inform.*, vol. 17, no. 5, pp. 3469–3477, May 2021.
- [4] J. Slay and M. Miller, "Lessons learned from the Maroochy water breach," in *Proc. Int. Conf. Crit. Infrastruct. Protection*, 2007, pp. 73–82.
- [5] S. M. Kasongo and Y. Sun, "A deep long short-term memory based classifier for wireless intrusion detection system," *ICT Exp.*, vol. 6, no. 2, pp. 98–103, 2020.
- [6] X. Zhou, W. Liang, S. Shimizu, J. Ma, and Q. Jin, "Siamese neural network based few-shot learning for anomaly detection in industrial cyber-physical systems," *IEEE Trans. Ind. Inform.*, vol. 17, no. 8, pp. 5790–5798, Aug. 2021.
- [7] S. Huang et al., "A gated few-shot learning model for anomaly detection," in *Proc. Int. Conf. Inf. Netw.*, 2020, pp. 505–509.
- [8] Y. Tian, Y. Wang, D. Krishnan, J. B. Tenenbaum, and P. Isola, "Rethinking few-shot image classification: A good embedding is all you need?," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 266–282.
- [9] J. Liu, L. Song, and Y. Qin, "Prototype rectification for few-shot learning," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 741–756.
- [10] O. A. Beg, T. T. Johnson, and A. Davoudi, "Detection of false-data injection attacks in cyber-physical DC microgrids," *IEEE Trans. Ind. Inform.*, vol. 13, no. 5, pp. 2693–2703, Oct. 2017.
- [11] Q. Sun, K. Zhang, and Y. Shi, "Resilient model predictive control of cyber-physical systems under DoS attacks," *IEEE Trans. Ind. Inform.*, vol. 16, no. 7, pp. 4920–4927, Jul. 2020.
- [12] F. Li, Y. Shi, A. Shinde, J. Ye, and W. Song, "Enhanced cyber-physical security in Internet of things through energy auditing," *IEEE Internet Things J.*, vol. 6, no. 3, pp. 5224–5231, Jun. 2019.
- [13] M. Kravchik and A. Shabtai, "Detecting cyber attacks in industrial control systems using convolutional neural networks," in *Proc. Workshop Cyber-Physical Syst. Secur. Privacy*, 2018, pp. 72–83.
- [14] N. Moustafa and J. Slay, "The evaluation of network anomaly detection systems: Statistical analysis of the UNSW-NB15 data set and the comparison with the KDD99 data set," *Inf. Secur. J., Glob. Perspective*, vol. 25, no. 1/3, pp. 18–31, 2016.
- [15] H. Pearce, S. Pinisetty, P. S. Roop, M. M. Y. Kuo, and A. Ukil, "Smart I/O modules for mitigating cyber-physical attacks on industrial control systems," *IEEE Trans. Ind. Inform.*, vol. 16, no. 7, pp. 4659–4669, Jul. 2020.
- [16] T. Huang, P. Chen, and R. Li, "A semi-supervised VAE based active anomaly detection framework in multivariate time series for online systems," in *Proc. ACM Web Conf.*, 2022, pp. 1797–1806, doi: 10.1145/3485447.3511984.
- [17] M. Zheng, J. Man, D. Wang, Y. Chen, Q. Li, and Y. Liu, "Semi-supervised multivariate time series anomaly detection for wind turbines using generator SCADA data," *Reliab. Eng. Syst. Saf.*, vol. 235, Jul. 2023, Art. no. 109235, doi: 10.1016/j.res.2023.109235.
- [18] H. Cho, J. Seol, and S.-G. Lee, "Masked contrastive learning for anomaly detection," in *Proc. 30th Int. Joint Conf. Artif. Intell.*, 2021, pp. 1434–1441.
- [19] R. D. Hjelm et al., "Learning deep representations by mutual information estimation and maximization," in *Proc. Int. Conf. Learn. Representations*, 2019, pp. 1–24.
- [20] O. Kopuklu, J. Zheng, H. Xu, and G. Rigoll, "Driver anomaly detection: A dataset and contrastive learning approach," in *Proc. IEEE Winter Conf. Appl. Comput. Vis.*, 2021, pp. 91–100.
- [21] T. Shenkar and L. Wolf, "Anomaly detection for tabular data with internal contrastive learning," in *Proc. 10th Int. Conf. Learn. Representations*, 2022, pp. 1–26.
- [22] Y. Wang, Q. Yao, J. T. Kwok, and L. M. Ni, "Generalizing from a few examples: A survey on few-shot learning," *ACM Comput. Surv.*, vol. 53, no. 3, 2020, Art. no. 63.
- [23] M. Uchida, "Human error tolerant anomaly detection based on time-periodic packet sampling," *Knowl.-Based Syst.*, vol. 106, pp. 242–250, 2016.
- [24] R. Kale and L. L. V. Thing, "Few-shot weakly supervised cybersecurity anomaly detection," *Comput. Secur.*, vol. 130, 2023, Art. no. 103194.
- [25] A. Puzanov, S. Zhang, and K. Cohen, "Deep reinforcement one-shot learning for artificially intelligent classification in expert aided systems," *Eng. Appl. Artif. Intell.*, vol. 91, 2020, Art. no. 103589.
- [26] P. Bedi, N. Gupta, and V. Jindal, "Siam-IDS: Handling class imbalance problem in intrusion detection systems using Siamese neural network," *Procedia Comput. Sci.*, vol. 171, pp. 780–789, 2020.
- [27] I. A. Khan, M. Keshk, D. Pi, N. Khan, Y. Hussain, and H. Soliman, "Enhancing IIoT networks protection: A robust security model for attack detection in Internet industrial control systems," *Ad Hoc Netw.*, vol. 134, 2022, Art. no. 102930.
- [28] P. Wu and H. Guo, "LuNET: A deep neural network for network intrusion detection," in *Proc. IEEE Symp. Ser. Comput. Intell.*, 2019, pp. 617–624.
- [29] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *Proc. Mil. Commun. Inf. Syst. Conf.*, 2015, pp. 1–6.
- [30] M. Tavallae, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *Proc. IEEE Symp. Comput. Intell. Secur. Defense Appl.*, 2009, pp. 1–6.
- [31] A. Alsaedi, N. Moustafa, Z. Tari, A. Mahmood, and A. Anwar, "TON_IoT telemetry dataset: A new generation dataset of IoT and IIoT for data-driven intrusion detection systems," *IEEE Access*, vol. 8, pp. 165130–165150, 2020.
- [32] S. E. Quincozes, C. Albuquerque, D. G. Passos, and D. Mossé, "ERENO: A framework for generating realistic IEC-61850 intrusion detection datasets for smart grids," *IEEE Trans. Depend. Secure Comput.*, pp. 1–15, 2023.



Haili Sun received the B.S. degree in information and computer science from Hainan University, Haikou, China, in 2013, and the M.S. degree in computer technology in 2015 from the School of Computer Science and Engineering, Huazhong University of Science and Technology, Wuhan, China, where she is currently working toward the Ph.D. degree in cyberspace security with the School of Cyber Security and Engineering.

Her research interests include anomaly detection and security control of industrial IoT systems, and theory and application of cyber-physical systems.



Hongle Liu received the B.S. degree in automation in 2022 from the Huazhong University of Science and Technology, Wuhan, China, where she is currently working toward the M.S. degree in cyber science and engineering with the School of Cyber Security and Engineering.

Her current research focuses on software security.



Yan Huang (Member, IEEE) received the M.S. and Ph.D. degrees in computer science and technology from the Huazhong University of Science and Technology, Wuhan, China, in 2016 and 2022, respectively.

Since 2022, he has been a Postdoctor with the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. His research interests include knowledge graph embedding, knowledge inference, and their application in object detection, cyber

security, and natural language processing.



Juan Chen received the B.S. degree in information and computational science from Northeastern University, Boston, MA, USA, in 2021. She is currently working toward the M.S. degree in cyberspace security with the School of Cyber Security and Engineering, Huazhong University of Science and Technology, Wuhan, China.

Her current research focuses on malicious behavior detection.



Chunjie Zhou received the M.S. and Ph.D. degrees in control theory and control engineering from the Huazhong University of Science and Technology, Wuhan, China, in 1991 and 2001, respectively.

He is currently a Professor with the School of Artificial Intelligence and Automation, Huazhong University of Science and Technology. His research interests include safety and security control of industrial control systems, theory and application of networked control systems, and

artificial intelligence.



Xin Li received the B.S. degree in cyber science and engineering in 2023 from the Huazhong University of Science and Technology, Wuhan, China, where he is currently working toward the M.S. degree in cyber science and engineering with the School of Cyber Science and Engineering.

His current research focuses on binary vulnerability inversion and Java application security.



Lansheng Han received the B.S. degree from Lanzhou University, Lanzhou, China, in 1995, and the Ph.D. degree from the Huazhong University of Science and Technology, Wuhan, China, in 2006, both in information security.

He is currently a Professor with the School of Cyber Security and Engineering, Huazhong University of Science and Technology, and the Peng Cheng Laboratory, Shenzhen, China. His research interests include computer virus, IoT security, malicious code detection, and mobile

intelligent terminal security.